Performance evaluation of a method for the detection and isolation of Shiga toxin producing *Escherichia coli* in foods

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1. INTRODUCTION

Shiga toxin *Escherichia coli* (STEC) are a group of pathogenic *E. coli* which are defined by the potential to produce one or more Shiga toxins (Stx, verotoxin) as defined by production of Stx, or presence of the gene *stx* (FAO/WHO 2018). STEC are also known as verotoxigenic producing *E. coli*. The term enterohemorrhagic *E. coli* is also used for STEC strains which share features with STEC O157:H7/NM (FAO/WHO 2018; Levine, 1987).

STEC have multiple evolutionary origins among *E. coli*, and so they are diverse in serotype, genotype, and physiology (Denamur et al., 2021; Geurtsen et al., 2022). STEC infections can result in illness ranging from uncomplicated diarrhea to bloody diarrhea, and life threatening haemolytic uremic syndrome (Joseph et al., 2020). STEC strains can be highly infectious, with a significant probability of infection on exposure to a single cell (Buvens et al., 2012; Teunis et al., 2008; Paton et al., 1996; Gill et al., 2019)

STEC illness is nationally notifiable disease in Canada, and the presence of this pathogen in food is regulated under the Food and Drug Act, and relevant Health Canada guidance documents. Regulatory and public health testing for STEC in foods is supported by methods published in The Compendium of Analytical Methods (GoC, 2025). The standard methods for the detection of STEC, including O157 and non-O157 serotypes is **MFLP-52 Isolation and identification of priority verotoxigenic *Escherichia coli* (STEC) in foods.** In the decade since MFLP-52 was published understanding of STEC has developed, and feedback from laboratories using MFLP-52 have identified opportunities for improvement. A primary consideration is that since MFLP-52 was originally published the range of STEC involved in human illness have been revealed to be more diverse (Gill et al., 2020). Most importantly when MFLP-52 was originally developed, Stx was distinguished simply as Stx1 and Stx2, since then a total of 19 subtypes of Stx (Stx1a,c,d,e, Stx2a to 2o) have been described with significant variability in protein and gene sequence which can make detection challenging (Bai et al., 2018; Bai et al., 2021; Gill et al., 2022; Lacher et al., 2016; Lindsey et al., 2023; Meng et al., 2014; Probert et al., 2014; Scheutz et al., 2012; Yang et al., 2022). Since the presence of *stx* or Stx is the essential diagnostic feature of STEC, there was a need to create a protocol which detected all known Stx subtypes.

MFLP-52 included characterisation of STEC isolates by a multiplex PCR and cloth-based hybridization array system (PCR-CHAS) for the identification of priority STEC based on the presence of genes for intimin, enterohemolysin, and O serogroups O157, O26, O103, O111 and O145. Since MFLP-52 was published in 2014, enterohemolysin and O-type have come to be viewed as having limited utility as markers of priority STEC (Gill et al., 2020). In parallel whole genome sequencing (WGS) has emerged as a technology for the routine characterisation of STEC. For these reasons this revised method no longer includes PCR-CHAS characterisation, or detection of O-type. Instead, this method includes real-time PCR for colonisation factors (*eae*, *aggR*, *aaiC*) associated with a greater risk of severe illness including HUS (FAO/WHO 2018, EFSA BIOHAZ Panel, 2020; Gill et al., 2020). These genes are markers for the presence of the locus of enterocyte effacement pathogenicity island (diagnostic gene, *eae*) and enteroaggregative attachment (diagnostic genes, *aaiC*/*aggR*) which involves both plasmid and chromosomaly encoded genes (FAO/WHO 2018, Gill et al., 2020). By rapidly identifying STEC isolates that carry these colonisation factors they may prioritised for follow up, including further characterisation by WGS or a new PCR-CHAS protocol (MFLP-119). However, STEC without these colonisation factors may still cause severe illness in vulnerable individuals, or present an increased risk of severe illness, such as strains producing Stx2a or Stx2d (EFSA BIOHAZ Panel, 2020; Lindqvist et al., 2023; Gill et al., 2020).

This report presents the results of absolute validation trials conducted to document the performance of a revised version of MFLP-52. The revision incorporates the following changes:

a. Changes to enrichment conditions to facilitate the recovery of injured STEC cells.

b. Changes to the isolation procedure to improve the probability of isolation in the presence of interfering microbiota.
c. Providing two options for DNA extraction to support analysis of food matrixes containing potential PCR inhibitors.
d. Replacement of conventional PCR detection of *stx* with a real-time PCR for *stx* (STX real-time PCR).
e. Addition of a real-time PCR for accessory virulence genes *eae*, *aaiC* and *aggR* which are associated with STEC that cause severe illness (CF real-time PCR).
f. The removal of PCR-CHAS isolate characterisation from the method and replacement with the option for further characterisation by WGS or an independent PCR-CHAS (MFLP-119).

2. PRINCIPLE OF ANALYSIS

Protocol Summary: Analytical units of food are suspended in Tryptic Soy broth with pyruvate (TSBp) at a 1:9 (weight:volume) ratio, and incubated for 3 hours at 42 °C. Selective agents vancomycin, cefsulodin, and bile salts are then added to the suspension and incubation is continued at 42 °C for a total of 16 to 24 hours. The enrichment broth is screened with the STX real-time PCR for *stx,* using primers and probes that detect 19 Shiga toxin subtypes (1a, 1c, 1d, 1e, 2a to 2o) (McMahon et al., 2024). Enrichment broth cultures positive for *stx* are plated onto Rainbow® agar O157 (RBA) and MacConkey agar with 2.5 mg/L potassium tellurite (MAC-T), which are incubated at 35 °C for 16 to 24 h. Colonies are screened batchwise in pools by the STX real-time PCR to identify potential STEC isolates. Isolates positive for *stx* are characterised with the CF real-time PCR for the presence of three genes (*eae*, *aaiC* and *aggR*) involved in host colonisation. Isolates positive with both the STX and CF real-time PCR are reported as confirmed STEC. Isolates positive for *stx* and negative for CF are tested for species identity, and if they are *E. coli* reported as confirmed STEC. Confirmed STEC isolates can undergo more detailed characterisation of the virulence profile by STEC CHAS analysis (validated separately and submitted to MMC as MFLP-119), orby whole genome sequence analysis. The overall scheme for the method is presented in **Figure 1.**

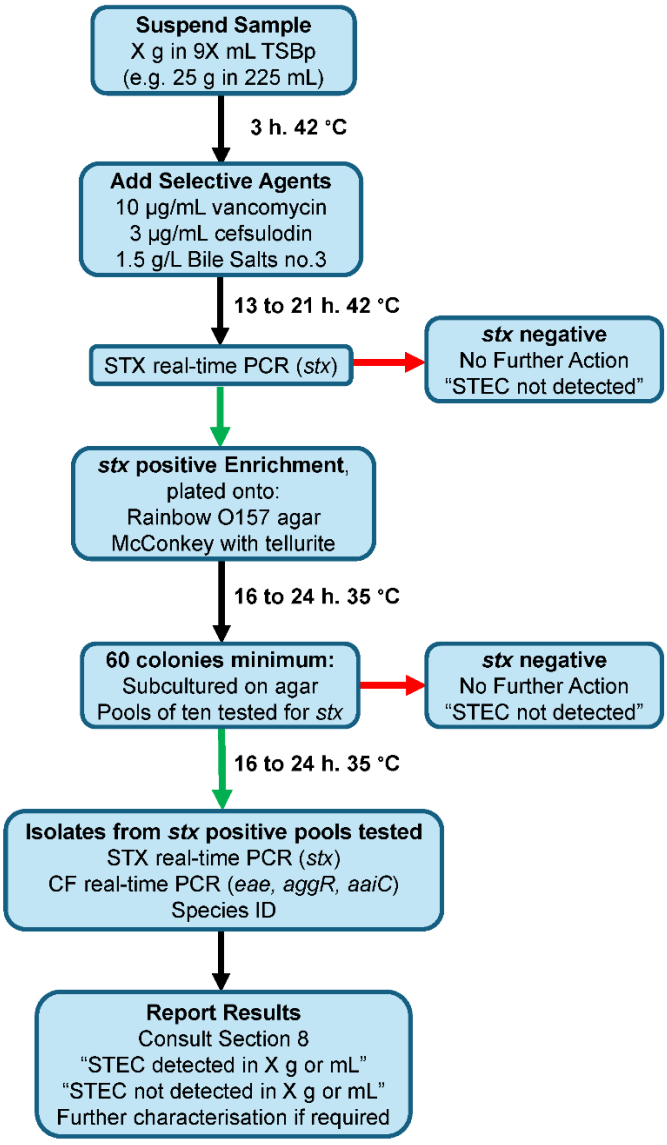


Figure 1. Flow diagram for analysis of food for Shiga toxin producing *Escherichia coli*

Recovery of injured STEC cells by enrichment. STEC are a subpopulation within the species *E. coli* with multiple phylogenetic origins, resulting in diverse growth characteristics and variable resistance to selective agents. The media, selective agents and growth conditions selected favour the recovery of diverse STEC. The use of 42 °C as an incubation temperature has been demonstrated to permit the replication of *E. coli*, while inhibiting some other bacteria (Gill et al., 2012). Antimicrobial agents (vancomycin, cefsulodin, and bile salts) are added to the enrichment media after 3 hours incubation to provide an opportunity for injured STEC cells to repair and start replicating, while providing a degree of selectivity against Gram positive bacteria, *Pseudomonas* spp. and *Proteus* spp. (Gill et al., 2012). Pyruvate is included in the enrichment media to aid the recovery of injured bacterial cells (Lee and Hartman, 1989). Pyruvate is a substrate of *E. coli* metabolism, and by degrading peroxide is protective against oxidative stress for cells with decreased catalase and superoxide dismutase activities (Gram et al., 1984). Experiments with three STEC strains determined that recovery of heat injured cells was maximal if bile salts no. 3, vancomycin and cefsulodin were added after 3 hours incubation in the base enrichment media at 42 °C (**SD-01**).

Post-enrichment DNA template extraction. To expand the scope of this method to different food types implicated in food safety events, the new method provides the option of using the **Puregene Cell Kit** (Qiagen), in addition to **InstaGene Matrix** (Bio-Rad). In experiments with sixteen different food matrices the Puregene Cell Kit provided more reliable analysis of six food matrices (soft cheese, cocoa, flaxseed, guacamole, peanut butter, walnut meat) than InstaGene Matrix (McMahon et al., 2023).

***Real-time PCR for stx* screening of enrichment and colonies.** To reduce the time to result and to streamline the screening procedures for enrichment and colony isolation, detection of *stx* in food enrichments, and bacterial colonies, uses the **STX real-time PCR.** The STX real-time PCR, which has been demonstrated to detect the genes for 19 subtypes (variants) of Stx by a multiplex reaction that uses a single probe label to report the presence of *stx,* without differentiating between *stx1* or *stx2*.

Colony Isolation. Isolation is attempted on two agar media with different selectivity profiles, RBA and MAC-T agar. RBA has low selectivity, permitting colony formation of all STEC tested, and other related bacteria (Gill et al., 2014). MAC-T is highly selective, in experiments MAC-T with 2.5 mg/L potassium tellurite was found to inhibit colony formation by 42.8% of STEC strains tested (n =200) and inhibit the growth of 88.5% of *stx* negative *E. coli* tested (n=200) **(SD-02**). Consequently, MAC-T may increase the probability of the isolation of tellurite resistant STEC in the presence of competing microbiota, including generic *E. coli.*

***Real-time PCR for* colonisation factors.** Three genetic markers *eae*, *aaiC*, and *aggR* are involved in host colonisation by some *E. coli* strains and are correlated with an increased risk of bloody diarrhea and hemolytic uremic syndrome, when present in STEC. The presence of one or more of those markers in presumptive STEC isolates is determined by the **CF real-time PCR**. Like the STX real-time PCR, the CF real-time PCR is a multiplex reaction with a single probe label. A positive result with the CF real-time PCR indicates that the STEC isolate poses an increased health risk, that should be prioritised for follow up. Isolates positive for *stx* and any of the colonisation factors are reported as STEC, as *eae*, *aaiC*, and *aggR* genes have not been reported present together with *stx* in bacteria other than *E. coli*, with exception of strains of the emerging pathogen *Escherichia albertii* carrying *stx* and *eae* (Gomes et al., 2020).

3. INCLUSIVITY AND EXCLUSIVITY OF POLYMERASE CHAIN REACTIONS

3.1 Inclusivity and Exclusivity STX real-time PCR

The STX real-time PCR is a multiplex protocol utilising nine primers, and four FAM labelled probes to detect DNA sequences for the allelic variants of the genes for 19 Stx subtypes (Stx1a,c,d,e, Stx2a to o). The reaction includes an internal positive control, with primers, probe (HEX labelled) and DNA template designed by Kavlick (2018). The development of the STX real-time PCR, including inclusivity and exclusivity studies is fully described in McMahon et al. (2024). The inclusivity and exclusivity of the STX real-time PCR for the detection of *stx* was assessed with the experiments described below. The *stx* gene can be found in bacteria other than *E. coli* and as free *stx*-phages and so the inclusivity and exclusivity panels were chosen to demonstrate specificity for *stx*, in all known allelic variants, rather than specificity for STEC.

Bacterial cultures stored at - 80 °C were plated onto one or more of the following media: Brain Heart Infusion (BHI) agar, MacConkey agar (MAC), MacConkey agar with 2.5 mg/L potassium tellurite (MAC-T), Rainbow® Agar O157 (RBA) (Biolog Inc.), Nutrient Agar (NA), and Tryptic Soy agar (TSA) or LB Agar with 50 µg/mL ampicillin (LBAA, transformants only) incubated for 16 to 24 h at 35 °C. The presence of Shiga toxin genes and the Stx subtypes present in all strains was determined by PCR with the protocol of Scheutz et al. (2012), or by sequence analysis (Gill et al., 2022). Bacterial strains carrying Stx subtypes 1e, 2k, 2l, 2m, and 2n were not available. To test the ability of the STX real-time PCR to detect the nucleotide sequences of these five Stx subtypes, the inclusivity panel included five *E. coli i*solates transformed with a plasmid encoding the nucleotide sequence for a toxoid based on one of subtypes Stx1e, Stx2k, Stx2l, Stx2m, or Stx2n (McMahon et al., 2024). The transformation of a bacterium with the sequence of known virulence factor would create a biosafety and biosecurity issue, and so the toxoid sequences include two-point mutations in the active site of the Stx A subunit, rendering it enzymatically inactive (McMahon et al., 2024). DNA template for PCR analysis was prepared by taking material from well separated single colonies and suspending in 100 µL of 1 % (v/v) Triton-X100, vortexed to mix, then heated at 100 °C for 10 min, before cooling at room temperature for 5 min. The lysate was used directly as template for the STX real-time PCR, analysed with a LightCycler 96 Instrument (Roche Diagnostics) or a CFX96 Instrument (BioRad).

The inclusivity panel consisted of a total of 160 bacteria with known Stx subtype sequences. These included 154 STEC strains (75 on BHI agar, 71 on MAC agar, 24 MAC-T agar, 24 RBA agar, 24 TSA and 24 NA), 1 strain of *stx*-positive *Shigella dysenteriae* (BHI agar), and 5 *E. coli* transformants carrying Stx toxoid sequences (LBAA) (**SD-03 Tab 1**). The inclusivity panel included at least one strain carrying a representative gene sequence for each Stx subtype, in the absence of other Stx subtypes. The exclusivity panel was composed of 70 *stx*-negative bacteria (62 on BHI agar, 8 on MAC) from 22 genera associated with food, clinical, and environmental samples, including 21 strains of *stx* negative *E. coli* (**SD-03 Tab 2**). The composition of the inclusivity and exclusivity panels is summarised in **Table 1**. All inclusivity panel strains were positive for *stx* with the STX real-time PCR. All exclusivity strains were negative for *stx* with the STX real-time PCR (**Table 2**).

Table 1. STX real-time PCR, Shiga toxin genotypes of inclusivity and exclusivity panel organisms.

Inclusivity Pane (n=160)			**Exclusivity Panel** (n=70)		
Organism	**Shiga toxin**	**Number of strains**	**Organism**	**Shiga toxin**	**Number of strains**
Escherichia coli	1a	50	*Bacillus spp.*	none	4
E. coli	1a,2a	11	*Citrobacter spp.*	none	2
E. coli	1a,2c	9	*Comamonas testosteroni*	none	1
E. coli	1c	4	*Escherichia coli*	none	21
E. coli	1c,2a	1	*Escherichia hermannii*	none	1
E. coli	1c,2b	3	*Enterobacter spp.*	none	4
E. coli	1c,2d	1	*Hafnia alvei*	none	1
E. coli	1d	6	*Klebsiella oxytoca*	none	2
E. coli	1e toxoid	1	*Kocuria varians*	none	1
E. coli	2a	20	*Listeria sp.*	none	7
E. coli	2a,2c	6	*Micrococcus flavus*	none	1
E. coli	2b	3	*Moellerella wisconsensis*	none	1
E. coli	2b,2c	1	*Morganella morganii*	none	1
E. coli	2b,2d	2	*Pasteurella pneumotropica*	none	1
E. coli	2b,2c,2d	1	*Proteus sp.*	none	2
E. coli	2c	8	*Pseudomonas sp.*	none	3
E. coli	2c,2d	1	*Raoultella ornithinolytica*	none	1
E. coli	2d	8	*Salmonella enterica*	none	8
E. coli	2e	5	*Serratia marcescens*	none	1
E. coli	2f	8	*Shigella spp*	none	3
E. coli	2g	2	*Staphylococcus spp.*	none	4
E. coli	2h	1			
E. coli	2i	1			
E. coli	2j	1			
E. coli	2k toxoid	1			
E. coli	2l toxoid	1			
E. coli	2m toxoid	1			
E. coli	2n toxoid	1			
E. coli	2o	1			
Shigella dysenteriae	*stx*	1			

Table 2. Performance parameters for the detection of *stx* in colony isolates by the STX real-time PCR:

True Positives	160	Sensitivity	100%
True Negatives	70	Specificity	100%
False Positives	0	False Negative Rate	0%
False Negatives	0	False Positive Rate	0%
Total	230		

3.1.1 *In Silico* Specificity of STX real-time PCR Primers and Probes

To supplement laboratory inclusivity and exclusivity experiments, *in silico* analysis was conducted to predict the specificity of the STX real-time PCR primers and probes for the detection of *stx* in over 27,000 bacterial genomes, including over 8,700 *Escherichia* genomes. The workflow of the *in-silico* analysis is presented in **Figure 2.** Custom python scripts, including the PrimerCheck script (https://github.com/bmh-vteclab/PrimerCheck), were used to predict the potential for primer/probe combinations to produce amplification products. The PrimerCheck script can predict primer and probe binding for the potential combinations of the multiple STX real-time PCR forward (5) primers, reverse (4) primers, and probes (4).

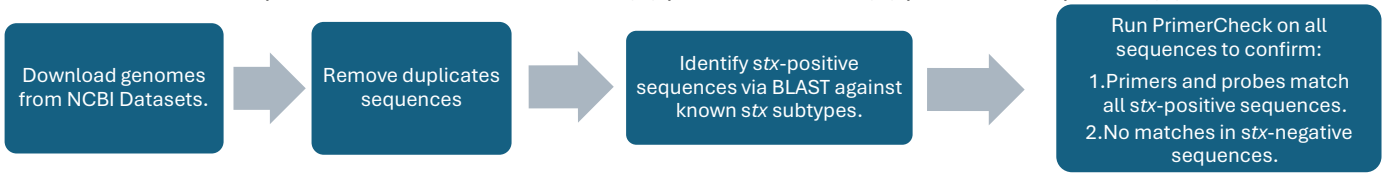


Figure 2. Workflow of *in silico* analysis

The PrimerCheck script predicts the potential for detection of amplification from the primers and probes by sequentially performing BLAST searches for different combinations of the forward primers and reverse primers and then matching probes to the predicted amplicons. This process includes steps to confirm that the forward and reverse primers bind to opposite strands; that a probe binds to a site between the forward and reverse primers; that primer and probe matches to amplicons have a minimum length of 17 nucleotides; that the percent sequence identity between binding sites and primers and probes is 88%; and that the product is less than 500 nucleotides long.

To validate the ability of the PrimerCheck script to correctly predict amplification the script was used to analyze nineteen sequences representing the different *stx* subtypes (positive controls) and 12 artificial test sequences representing potentially scenarios for both correct and incorrect binding of primers and probes (**SD-04 Tab 1**). The PrimerCheck script generated expected results with all test sequences.

For *in silico* inclusivity and exclusivity analysis of the STX real-time PCR with the PrimerCheck script, complete bacterial genomes were downloaded from NCBI of *Escherichia* and 29 other genera of Enterobacteriaceae, including *Enterobacter, Shigella, Citrobacter, Klebsiella, Atlantibacter, Salmonella*. Enterobacteriaceae genomes were selected as the enrichment and isolation media used is selective for the growth of Enterobacteriaceae, the family to which *E. coli* belongs. The resulting set of sequences included both chromosomal and plasmid sequences. The NCBI dataset included sequences assigned both Genbank and NCBI Reference Sequence accession numbers. When NCBI Reference sequences were determined to be duplicates, only the Genbank sequences were retained.

To identify genomes containing the *stx* gene, BLAST analysis (BLAST_genes script: https://github.com/bmh-vteclab/BLAST_Genes) was conducted of the database of bacterial sequences against type sequences for the 19 *stx* subtypes (**SD-04 Tab 1**), with an 88% percent identity cutoff. The BLAST analysis identified *stx*-positive sequences from 697 *Escherichia coli* sequences, 15 *E. albertii* sequences, 4 *Escherichia* sequences of unspecified species, 15 *Shigella* sequences and 1 *Enterobacter* sequence (**SD-04 Tab 3**). A total of 15,437 sequences of *Escherichia* (**SD-04 Tab 5**), and 31,132 sequences of other bacteria (**SD-04 Tab 6**) were *stx*-negative. The PrimerCheck script was run against all the sequences. The PrimerCheck script was also run against 123 *stx*-positive bacteria (*E. coli* 118, *E. albertii* 1, *Enterobacter* 2, *Citrobacter* 1, *Acinetobacter* 1), strains and 14 *stx*-phages of known subtypes to confirm that the STX real-time PCR detects all known subtypes (**SD-04 Tab 2 and Tab 4**).

The PrimerCheck determined that all the *stx*-positive sequences from bacteria and bacteriophage showed strong matches (>90% sequence identity) for at least one forward primer, one reverse primer, and one probe of the STX real-time PCR (**SD-04 Tab 2 and Tab 3**). The STX real-time PCR was not predicted to produce amplification products from any of the *stx*-negative sequences. This provides strong evidence for the specificity of the STX real-time PCR primers and probes, confirming their ability to detect all known *stx* subtypes, without producing false positives from *stx*-negative sequences (**SD-04 Tab 5, Tab 6**).

3.1.2 Interlaboratory Study of STX real-time PCR

To demonstrate the reliability of the STX real-time PCR when conducted by different laboratories using different real-time PCR platforms, microwell test plates were prepared with DNA extracts from *stx* positive and negative *E. coli* strains. The test plates were distributed to four CFIA laboratories (Dartmouth, Greater Toronto Area, Ottawa Laboratory Carling, St-Hyacinthe) and the Health Canada Microbiology Laboratory Longueuil, for analysis on four real-time PCR platforms (Roche Lightcycler 96, BioRad CFX96, Applied Biosystems QuantStudio 5 and 6,).

Genomic DNA was extracted from cultures of five STEC and one *stx* negative strain of *E. coli* (**Table 3; SD-05 Tab 1**) using the Maxwell RSC Cultured Cells DNA Kit (Promega). The extracted DNA was quantified using the Qubit dsDNA HS Assay Kit (Invitrogen). The quantified DNA was diluted in PCR grade water to a final concentration of 5 ng/µL. For each DNA extract, triplicate wells on the test plate were loaded with 5 µL of the diluted DNA, resulting in a final concentration of 25 ng per well (**SD-05 Tab 1**). The plates were then allowed to dry overnight in biosafety cabinet. Once dried, they were sealed and stored at 4 °C.

Table 3. *Escherichia coli* strains used to prepare DNA template for interlaboratory trial.

Strain ID (OLCID)	Source Matrix	O-type	Genotype
OLC 455	Unknown	O111	*stx1a, eae*
OLC 710	Unknown	O121	*stx2a, eae*
OLC 716	Unknown	O45	*stx1a, eae*
OLC 797	Human	O157	*stx1a, stx2a, eae*
OLC 684	Unknown	O145	*eae*
OLC 2346	Raw Ground Pork	O157	None (negative control)

The plates were shipped to laboratories where the DNA template was recovered for PCR by rehydrating each well with 25 µL of PCR-grade water, to achieve a final concentration of 1 ng/µL. The plate was vortexed and spun down to ensure complete rehydration. STX real-time PCR analysis was then conducted, with 2.5 µL of the rehydrated gDNA as the template and 22.5 µL of real-time PCR mastermix, for a total reaction volume of 25 µL. The results from individual labs are presented in **SD-05** (**Tabs 3 to 9**), and summarised in (**Table 4, SD-05 Tab 2)**.

Table 4. Results of interlaboratory study of STX real-time PCR.

Laboratory	Platform	True Positive	True Negative	False Positive	False Negative	Notes
OLC	Lightcycler 96	12	7	0	0	
Dartmouth	Lightcycler 96	12	7	0	0	
St-Hyacinthe	Lightcycler 96	12	7	0	0	
GTA	Lightcycler 96	11	7	0	1	
Longueuil	CFX96	11	7	0	0	Sample C1 lost in transit
Longueuil	Quantstudio 5	11	7	0	0	Sample C1 lost in transit
OLC	Quantstudio 6	12	7	0	0	
Total		81	49	0	1	

Sensitivity 98.8%; Specificity 100%; False Negative Rate 1.2%; False Positive Rate 0.0%; Agreement 99.2%

3.2 Inclusivity and Exclusivity CF real-time PCR

The CF (colonisation factor) real-time PCR is a multiplex protocol utilising 6 primers and 3 FAM labelled probes to detect DNA sequences for *eae, aggR and aaiC*. The reaction includes an internal positive control (IPC) primers, probes and DNA template designed by Kavlick (2018). The inclusivity and exclusivity of the CF real-time PCR for the detection of *eae, aggR and aaiC* was assessed with the experiments described below. The CF real-time PCR targets three genes associated with colonisation of human hosts by STEC, but these genes are not unique to STEC, nor are they carried by all STEC. Consequently, the inclusivity and exclusivity panels were chosen to demonstrate specificity for *eae, aggR and aaiC* rather than specificity for STEC.

Cultures stored at - 80 °C were plated onto one or more of the agar media BHI, MAC, MAC-T, RBA, TSA, and NA, and incubated for 16 to 24 h at 35 °C. The presence of *eae, aggR, aaiC* in *E. coli* strains was determined by genome sequencing. DNA template for PCR analysis was prepared by taking material from well separated single colonies and suspending in 100 µL of 1 % (v/v) Triton-X100, vortexed to mix, then heated at 100 °C for 10 min, before cooling at room temperature for 5 min. The lysate was used directly as template for the CF real-time PCR, analysed with a LightCycler 96 Instrument (Roche Diagnostics) or a CFX96 Instrument (BioRad).

The inclusivity panel consisted of a total of 118 *E. coli* strains that carry one or more of the colonisation factors *eae*, *aggR* or *aaiC* (47 on BHI agar, 61 on MAC agar, 22 on MAC-T agar, 22 on RBA agar, 22 on TSA and 22 on NA) (**SD-03 Tab 3**). The genotypes included one strain carrying *aaiC*, seven strains carrying *aaiC* and *aggR*, and 99 strains carrying *eae*. The exclusivity panel was composed of 72 bacteria negative for the three colonisation factors (67 on BHI agar, 5 on MAC, 2 on MAC-T, 2 on RBA, 2 on TSA and 2 on NA) (**SD-03 Tab 4**). The bacteria included single strains of *Bacillus licheniformis, B. sphaericus, Citrobacter braakii, Comamonas testosteroni, Enterobacter kobei, Escherichia hermannii, Hafnia alvei, Klebsiella oxytoca, Kocuria varians, Listeria monocytogenes, Moellerella wisconsensis, Morganella morganii, Pasteurella pneumotropica, Proteus mirabilis, Pseudomonas fluorescens, Ps. stutzeri, Raoultella ornithinolytica, Shigella dysenteriae, Sh. flexneri, Staphylococcus sciuri,* three strains of *Salmonella enteritidis* and 49 strains of *E. coli.* All inclusivity panel strains were positive with the CF real-time PCR. All exclusivity strains were negative with the CF real-time PCR (**Table 5**).

Table 5. Performance parameters for the detection of *aggR, aaiC* or *eae* in colony isolates by CF real-time PCR.

True Positives	118	Sensitivity	100%
True Negatives	73	Specificity	100%
False Positives	0	False Negative Rate	0%
False Negatives	0	False Positive Rate	0%
Total	191		

3.2.1 *In Silico* Specificity of CF real-time PCR Primers and Probes

To supplement laboratory inclusivity and exclusivity experiments, *in silico* analysis was conducted to predict the specificity of the CF real-time PCR primers and probes for the detection of *eae, aaiC* and *aggR* in over 27,000 bacterial genomes, including over 8,700 *Escherichia* genomes. This analysis was conducted on the same set of sequences (*Escherichia* and 29 other genera of Enterobacteriaceae, including *Enterobacter, Shigella, Citrobacter, Klebsiella, Atlantibacter, Salmonella*) and with the same workflow and tools as described in Section 3.1.1. Type sequences for *aaiC*, *aggR*, and 21 variants of *eae* are listed in **SD-06 Tab 1**. The PrimerCheck script generated expected results with all test sequences with exception of *eae*-citrobacter, a variant of *eae* found in *Citrobacter* species.

To identify genomes containing the *eae, aaiC* or *aggR* (CF genes), BLAST analysis (BLAST_Genes script) was conducted of the database of bacterial sequences against type sequences for the 23 type sequences (**SD-06 Tab**

1), with an 88% percent identity cutoff. The BLAST analysis identified 761 *E. coli*, 87 *E. albertii*, 4 *Citrobacter rodentium* and 2 *Shigella flexneri* sequences containing a CF gene (**SD-06 Tab 3**). A total of 15,305 sequences of *Escherichia* (**SD-06 Tab 4**)*,* and 31,126 sequences of other bacteria (**SD-06 Tab 5**) were negative for the CF genes. The PrimerCheck script was run against all the sequences. The PrimerCheck script was also run against the sequences of 68 CF gene-positive bacteria (*E. coli* 43, *E. albertii* 21, *Escherichia sp. 1, Citrobacter freundii* 1, *Shigella boydii* 2) to confirm that the CF real-time PCR detects all known variants of the three CF genes, the only gene variant that was not detected was *eae*-citrobacter (**SD-06 Tab 2**).

PrimerCheck showed that, with the exception of *eae*-positive *Citrobacter* strains, all the CF-positive sequences from bacteria and bacteriophage exhibited strong matches (>90% sequence identity) to at least one forward primer, one reverse primer, and one probe used in the CF real-time PCR (**SD-06 Tab 2, Tab 3**). The CF real-time PCR was not predicted to produce amplification products from any of the CF-negative sequences. This provides strong evidence for the specificity of the CF real-time PCR primers and probes, confirming their ability to detect all known *eae* subtypes found in *E. coli*, as well as *aaiC* and *aggR,* without producing false positives from sequences that lack the three CF genes.

3.2.2 Trial of CF real-time PCR on multiple PCR platforms

To demonstrate the reliability of the CF real-time PCR when conducted using different real-time PCR platforms, microwell test plates were prepared with DNA extracts from four *E. coli* positive for the CF genes (*aaiC*, *aggR*, *eae*) and two *E. coli* strains negative for the CF genes. The test plates were analysed on three real-time PCR platforms (Roche Lightcycler 96, BioRad CFX96, Applied Biosystems QuantStudio 6).

Genomic DNA was extracted from cultures of six *E. coli* strains (**Table 6**) using the Puregene Cell kit (Qiagen). The extracted DNA was quantified using the Qubit dsDNA HS Assay Kit (Invitrogen). The quantified DNA was diluted in PCR grade water to final concentration of 1 ng/µL using PCR-grade water. CF real-time PCR analysis was conducted, with 2.5 µL of the gDNA as the template and 22.5 µL of qPCR mastermix, for a total reaction volume of 25 µL. The results are presented in **SD-07** and summarised in **Table 7**.

Table 6. *Escherichia coli* strains used to prepare DNA template for real-time PCR platform study

Strain ID	Source Matrix	O-Type	Virulence Genes
11-3088	Human	O104	*stx2a, aaiC*
02-2862	Human	O86	*aggR, aaiC*
ATCC 35150	Human	O157	*stx1a, stx2a, eae*
PARC 443	Unknown	O157	*eae*
BMH-19-0006	Raw milk cheese	O91	*stx1a, stx2a, stx2d*
BMH-17-0032	Flour	O124	

Table 7. Results of CF real-time PCR platform study

Laboratory	Platform	True Positive	True Negative	False Positive	False Negative
OLC	Lightcycler 96	12	7	0	0
Health Canada	CFX96	12	7	0	0
OLC	Quantstudio 6	12	7	0	0
Total		36	21	0	0

Sensitivity 100%; Specificity 100%; False Negative Rate 0.0%; False Positive Rate 0.0%; Agreement 100%

4. EXPERIMENTS WITH INOCULATED FOOD

4.1 Bacterial Strains Inoculated to Foods

Details of the strains of *Escherichia coli* used for the inoculation of food samples for the validation study are presented in **Table 8.** The strains included STEC, and *stx* negative *E. coli,* which were added to selected food items as competing organisms. The strains were selected based on their isolation source, diversity of virulence gene profile, and serotype. A broad range of STEC strains was included, reflecting the method's scope to detect all STEC, not just those belonging to previously identified 'priority' serogroups.

Table 8. *Escherichia coli* strains inoculated to food samples.

Strain ID	Serotype	Shiga toxin profile	Colonisation factors	Isolation source	BMH Database Sequence
OLC 3602	OUT:H21	*stx2i*	None	Lamb	MER-2019-005470
LAAFC O103-2	O103:H2	*stx1a*	*eae*	Beef	MER-2024-012114
11-3088	O104:H4	*stx2a*	*aaiC*	Clinical (sprouts outbreak)	MER-2024-012113
OLC 1056	O170:H18	*stx2h*	None	Sprouts	CP076706*
BMH-19-0006	O91:H21	*stx1a, stx2a, stx2d*	None	Raw milk cheese	BMH-2019-001324
BMH-13-0001	O157:H7	*stx1a, stx2a*	*eae*	Raw milk cheese	BMH-2023-000277
1934	O157:H7	*stx1a, stx2a*	*eae*	Beef	BMH-2025-000348
BMH-22-0001	O1:H20	None	None	Kimchi	BMH-2023-000644
FD-51	O13/O129:H11	None	None	Spinach	BMH-2025-000466
OLC 1214	O148:H53	None	None	Ground beef	MER-2024-012210

OUT – O untypable; *GenBank

4.2 Method Performance: Inoculation of Foods and Enumeration of Microbiota

The selection of Food Categories and Food Types was informed by the Compendium of Analytical Methods (CAM) Vol. 1, Table 4.1: Classification of food categories and suggested food type-pathogen combinations for validation studies. Experiments included three Food Categories Dairy (**Table 9**), Fruit and Vegetable Based Products (**Table 10**), and Raw Meat Products (**Table 11**). Each Food Category was represented by three Food Types, with two Food Items for each Food Type.

Bacterial cultures stored at -80 °C as glycerol stocks were plated onto BHI agar or Tryptic Soy agar (TSA), and incubated for 18 to 24 h at 35 °C. A single colony of the selected culture was inoculated to a test tube of BHI broth which was incubated for 16 to 24 h at 35 °C. The cells were pelleted by centrifugation, the supernatant was discarded and the cell pellet resuspended in phosphate buffered saline (PBS), this process was repeated and the final cells suspension adjusted to an optical density of 0.10 at 600 nm, to provide a density of approximately 7.5 log colony forming units (CFU)/mL. For STEC inoculation, the cell suspension was diluted in PBS to a concentration that when inoculated as a 100 µL volume would give a low inoculum (fractional positives, <5 CFU/analytical unit) or a high inoculum (20 to 50 CFU/analytical unit). The inoculum was enumerated by plating 20 aliquots of the inoculum volume onto individual plates of TSA, that were incubated 16 to 24 h at 35 °C. The colonies on the TSA plates were counted and the average inoculum calculated as CFU/analytical unit. One food type in each category (Dairy – Raw; Fruit and Vegetable Based Products – Raw and Raw Processed; Raw Meats – Processed Meats) was also co-inoculated with a *stx* negative *E. coli* strain at 10 x the concentration of the STEC.

For each experiment 25 analytical units, divided between two food items of the same food type, were weighed into sampling bags. Twenty of the analytical units were inoculated with 100 µL of inoculum, seven analytical units were left uninoculated of which five were used as negative controls. The two remaining uninoculated analytical units were prepared for enumerating microbiota. The analytical units were then equilibrated at 2 to 8 °C for 3 days, with the exception of Food Type Nuts/Nut Meats which were equilibrated at room temperature for 17 to 18 days, and Food Type Frozen Meats which were equilibrated at -20 °C for 17 days. Once the equilibration period was completed the twenty inoculated analytical units and five of the uninoculated analytical units were analysed in parallel for STEC. The two remaining analytical units were assayed for total aerobic plate count by MFHPB-18 *Determination of the Aerobic Colony Counts in Foods* (GoC, 2025; Vol 2).

Table 9. Food Category Dairy.

Food Type	Food Item	Microbiota log CFU/g	Analytical Unit	Inoculated STEC		Co-inoculum		Storage	Enrichment Duration
				Strain	CFU	Strain	CFU		
Fermented	Yogurt	7.8	25 g	BMH-19-0006 (*stx1a, 2a, 2d*)	0.9	ND	ND	2-8 °C 3 days	24 h.
	Kefir	4.5							
	Yogurt	8.4			9.9				18 h.
	Kefir	4.0							
Heat Processed	Milk 2%	1.6	25 g	BMH-13-0001 (*stx1a, 2a, eae*)	0.8	ND		2-8 °C 3 days	19 h.
	Cheese Curds	8.7							
	Milk 2%	1.8			8.4				24 h.
	Cheese Curds	6.8							
Raw	Alfred le Fermier	5.7	25 g	BMH-13-0001 (*stx1a, 2a, eae*)	5.2	BMH-22-0001	44.5	2-8 °C 3 days	18 h.
	Brie de Meaux	8.7							
	Alfred le Fermier	7.7			9.0		94.4		24 h.
	Brie de Meaux	8.6							

Co-inoculum: *stx* negative E. coli strain added as interfering organism.
ND: Not Done.

Table 10. Food Category Fruit and Vegetable Based Products.

Food Type	Food Item	Microbiota log CFU/g	Analytical Unit	Inoculated STEC		Co-inoculum		Storage	Enrichment Duration
				Strain	CFU	Strain	CFU		
Nuts/Nut Meat	Peanut Butter	1.5	25 g	OLC 1056 (*stx2h*)	27.2	ND	ND	Room temperature 17 to 18 days	24 h.
	Walnuts	2.8							
	Peanut Butter	1.7			57.0				21 h.
	Walnuts	1.5							
Raw and Raw Processed	Romaine Hearts	3.1	25 g	11-3088 (*stx2a, aaiC)*	0.4	FD-51	9.8	2-8 °C 3 days	24 h.
	Baby Spinach	6.9							
	Romaine Hearts	6.5			4.5		41.4		24 h.
	Baby Spinach	7.8							
Juices	Vegetable Juice	3.8	25 g	OLC 1056 (*stx2h*)	0.8	ND	ND	2-8 °C 3 days	18 h.
	Apple Cider	0.7							
	Vegetable Juice	4.1			10				24 h.
	Apple Cider	0.7							

Co-inoculum: *stx* negative *E. coli* strain added as interfering organism.
ND: Not Done.

Table 11. Food Category Raw Meat Products.

Food Type	Food Item	Microbiota log CFU/g	Analytical Unit	Inoculated STEC		Co-inoculum		Storage	Enrichment Duration
				Strain	CFU	Strain	CFU		
Unprocessed	Beef Blade Roast	5.8	65 g	OLC 3602 (*stx2i*)	0.9	ND	ND	2-8 °C 3 days	24 h.
	Pork Loin	3.6							
	Beef Blade Roast	6.6			8.5				18 h.
	Pork Loin	6.9							
Processed	Ground Beef	7.7	65 g	LAAFC O103-2 (*stx1a*, *eae)*	1.4	OLC 1214	13.5	2-8 °C 3 days	24 h.
	Ground Pork	6.6							
	Ground Beef	7.3			14.1		182.2		18 h.
	Ground Pork	7.3							
Frozen	Beef Burger	2.6	65 g	LAAFC O103-2 (*stx1a*, *eae)*	0.8	ND	ND	-20 °C 17 days	24 h.
	Pork Burger	4.4							
	Beef Slices	<0.7			9.3				18 h.
	Pork Burgers	2.0							

Co-inoculum: *stx* negative *E. coli* strain added as interfering organism.
ND: Not Done.

4.3 Method Performance: Food Item Analysis

Due to the absence of a standard method that is inclusive for all STEC, this validation study was conducted as an Absolute Validation, with probability of detection (POD) assessed against the presence of STEC in the inoculated food items, determined by detection of *stx* in the enrichment of artificially inoculated analytical units.

For each experiment 25 analytical units were prepared, twenty (20) inoculated and five (5) uninoculated, were analysed in parallel according to the protocol **SD-A.** The protocol is summarised below.

Tryptic Soy broth with pyruvate (TSBp) was prewarmed to 42 °C. Analytical units of 25 g or 25 mL were suspended in 225 mL of TSBp (Categories Dairy and Fruit and Vegetable Products). Analytical units of 65 g were suspended in 585 mL of TSBp (Category Raw Meat Products). The suspensions were stomached and then incubated static for 3 hours at 42 °C. Selective agents, vancomycin (final concentration 10 µg/mL) , cefsulodin (3 µg/mL final concentration) and bile salts no. 3 (1.5 g/L final concentration) were added to the food suspensions and the incubation at 42 °C continued to a minimum of 18 hours and a maximum of 24 hours (Tables 9, 10 and 11).

After completion of the incubation two 1.5 mL aliquots of enrichment broth were transferred to sterile 2 mL microcentrifuge tubes and centrifuged at 500 x g for 1 minute to pellet food matrix particles. A 1 mL aliquot of the aqueous phase of the supernatant was then processed with InstaGene Matrix (BioRad) and the Puregene Cell Kit (Qiagen) to prepare two DNA templates. In routine operation the analyst has the choice of using either InstaGene Matrix or Puregene Cell Kit as appropriate for the food matrix. Both DNA extraction methods were included in parallel in the experiment to allow their relative performance to be assessed. Both InstaGene Matrix and Puregene Cell Kit DNA templates were screened for the presence of *stx* with the STX real-time PCR. Once STX real-time PCR screening was completed, aliquots of the enrichments were serially diluted in 0.1% peptone water and the 100 µL of dilutions 10^{-4}, 10^{-5} and 10^{-6} plated onto Rainbow® Agar O157 (RBA) (Biolog Inc.) and MacConkey agar with 2.5 mg/L potassium tellurite (MAC-T). The agar plates were incubated at 35 °C for 16 to 24 hours.

The morphology of STEC varies significantly between strains. Recovery of STEC isolates was attempted from the agar plates from *stx* positive enrichments by “pick and pool” of randomly selected colonies. Material from individual colonies on the agar plates was used to inoculate a BHI or nutrient agar plate with numbered grid, and then remaining colony material added to 100 µL of 1% (v/v) Triton X-100 in a sterile 1.5 mL screw-cap microcentrifuge tube. This process was repeated to create pools of ten colonies, with up to six (6) pools (a total of sixty colonies) screened per sample. The inoculated non-selective agar plates were incubated for 16-24 hours

at 35 °C. During incubation, DNA templates for each pool were prepared by heating at 100°C for 10 minutes, followed by cooling at room temperature for 5 minutes. The pools were then screened with the STX real-time PCR assay. If a pool tested positive for *stx,* DNA templates from the ten individual isolates within that pool were screened with the STX real-time PCR. Isolates that tested positive for *stx* were further screened using the CF real-time PCR assay. Isolates which tested positive for *stx* and CF real-time PCR were identified as STEC. Isolates which tested positive for *stx* and were negative by CF real-time PCR underwent species identification with VITEK (bioMérieux), if these isolates were identified as *Escherichia coli*, the isolate was identified as STEC. Analytical units were reported as STEC positive if at least one STEC isolate was recovered.

4.3.1 Confirmation of Results by Colony Sweeps

Isolating STEC on agar media in the presence of other microbiota is challenging, as the presence of *stx* is the only differential characteristic known to be unique to STEC, and no selective characteristics unique to STEC and not shared with other *E. coli* have been identified. Since STEC colonies cannot be distinguished by morphology on agar media, colonies are randomly selected to construct pools that are then screened for the presence of *stx*. In the presence of abundant background microbiota, which is common on RBA due to its low selectivity, there is a risk that even when STEC are present, no STEC colonies will be selected. This risk will be inversely proportional to the relative concentration of STEC in the post-enrichment microbiota to other bacteria which can replicate on RBA. To determine if STEC were present in the enrichment but had not been isolated from the colony pools, we performed additional confirmation testing for all analytical units from which no positive isolate was obtained, irrespective of the PCR screening result, including uninoculated samples.

In this study the definitive indicator of the presence or absence of STEC on the agar plates was screening a Plate Sweeping with STX real-time PCR. The purpose of the Plate Sweep is to screen all the colonies present on a plate in a single step. Plates of the lowest enrichment dilution plated were selected, usually 10^{-4} of RBA and 10^{-1} of MAC-T.

To confirm the presence or absence of STEC on agar plates, DNA template was extracted from colony material on the plate, the “plate sweep”. The agar surface was rinsed repeatedly with 1 mL of 0.1% peptone water to suspend colony material. The suspension (approximately 500 µL) was recovered, heated at 100 °C for 20 min, cooled, centrifuged for 1 minute at 16 000 x *g* to pellet the cell debris and a 2.5 µL aliquot of supernatant used as template for the STX real-time PCR. A positive STX real-time PCR signal from a plate sweep was interpreted as indicating the presence of STEC. A negative PCR result from plate sweep was interpreted as evidence that the analytical unit was true negative. **Table. 12** lists the potential outcomes and the interpretation of final status of the analysis in terms of true sample status, PCR screen performance and cultural isolation performance. Note that it is not possible for the isolation procedure to produce a false positive result because isolation requires physical recovery of a viable organism followed by confirmation of *stx.* An isolate cannot yield a positive *stx* result unless STEC is truly present (**SD-08 to SD-16**).

Table 12. Interpretation of analysis final status.

stx screen	Isolation	Plate Sweep		Final Status		
		RBA 10^{-4}	MAC-T 10^{-1}	Sample	PCR	Isolation
-	-	-	-	Negative	**TN**	**TN**
-	-	-	+	Positive	**FN**	**FN**
-	-	+	-	Positive	**FN**	**FN**
-	-	+	+	Positive	**FN**	**FN**
+	-	-	-	Negative	**FP**	**TN**
+	-	-	+	Positive	**TP**	**FN**
+	-	+	-	Positive	**TP**	**FN**
+	-	+	+	Positive	**TP**	**FN**
-	+	Not done	Not done	Positive	**FN**	**TP**
+	+	Not done	Not done	Positive	**TP**	**TP**

TP: true positive. **TN**: true negative. **FN**: false negative. **FP**: false positive

4.4 Limit of Detection Experiments

To determine the limit of detection (LOD) of the STEC method, experiments were conducted in accordance with the procedure described in CAM. Vol. 1, Annex 4.5. *Determination of the Limit of Detection*.

LOD experiments were conducted with one food item each from the Food Categories Dairy, Fruit and Vegetable Based Products and Raw Meat (**Table. 13**). Commercially sourced Religieuse Cheese (raw milk, semisoft) and Romaine lettuce were used in the LOD experiments with analytical units of 100 g. Romaine lettuce was chopped with a sterile knife and then portioned into stomacher bags. The cheese was grated with a sterile grater and then portioned into stomacher bags.

Two attempts were made to conduct LOD experiments with commercially sourced fresh ground beef with 325 g analytical units. These experiments could not provide results due to interference by the presence of naturally contaminating STEC. To provide STEC free ground beef for the LOD experiment ground beef was prepared from commercially sourced intact beef cuts. Two Outside Round roasts were purchased, cut into approximately 20 cm cubes, which were placed in boiling water for 1 min. The beef was removed from the water and the cooked meat excised with sterile knife. The raw meat was then ground through a bench top grinder. The ground meat was mixed in a bowl and then portioned out into 325 g analytical units. The resulting ground beef had a total aerobic colony count of 4.5 log CFU/g.

Cultures stored at -80 °C were plated onto BHI agar and incubated for 16 to 24 h at 35 °C. A single colony of the selected culture was inoculated to a test tube of BHI broth which was incubated for 16 to 24 h at 35 °C. The cells were pelleted by centrifugation, the supernatant was discarded and the cell pellet resuspended in phosphate buffered saline (PBS), this process was repeated and the final cells suspension adjusted to an optical density of 0.10 at 600 nm to provide a density of approximately 7.5 log colony forming units (CFU)/mL. The cell suspension was diluted in PBS to a concentration that when inoculated as a 100 µL volume would give an inoculum between 1 and 5 CFU, and designated Inoculum 1. An aliquot of Inoculum 1 was serially diluted 1:1 in PBS to prepared dilutions from 1/2, 1/4, 1/8, 1/16 to a maximum of 1/32. This series of diluted inoculums were used to inoculate analytical units in stomacher bags, with 6 replicates for each dilution. The inoculum was massaged into the food item. To ensure that the microbiota of the ground beef included generic *E. coli* as competing organism, the ground beef analytical units were also co-inoculated with 40 CFU of *E. coli* OLC 1214 a ground beef isolate. The inoculated food was stored for 3 days at 1 to 5 °C to equilibrate. Following equilibration, the analytical units were analysed in parallel for STEC as described in Section 4.5 above.

The highest level of inoculum (Inoculum 1) was enumerated at the time of inoculation by immediately plating six 100 µL aliquots on TSA, which were incubated for 2 days at 35 °C. Following the requirements of CAM. Vol. 1, Annex 4.5. the highest inoculum in the food item at the time of analysis was also estimated by Most Probable Number (MPN) of three replicate 50 g portions of the food item. These 50 g portions were inoculated with 100 µL aliquots of Inoculum 1 and equilibrated in parallel with the other analytical units. On the day of LOD analysis each 50 g portion was suspended in 450 mL of TSBp and stomached. Aliquots from these suspensions were then diluted in TSBp to prepare triplicate sets of MPN tubes representing three different sample sizes: 45 g, 4.5 g and 0.5 g of the food item. The MPN tubes were then analysed for STEC as described in Section 4.5 above. The background microbiota of the food item was determined by analysis of three analytical units of 10 g for total aerobic plate count by MFHPB-18 (GoC, 2025).

Table 13. Limit of Detection Experiments. Food Item, microbiota levels, analytical unit, Inoculated STEC strain, enumeration of STEC inoculum by agar plating, and estimate of STEC inoculum by most probable number.

Food Item	Microbiota Log CFU/g	Analytical Unit (g)	Inoculated STEC	Highest Inoculum		Generic *E. coli* Co-inoculated (CFU)
				CFU	MPN (CL)	
Raw Milk Cheese	8.50	100 g	BMH-13-0001 *stx1a, stx2a eae*	2.3	1.65 (0.45-5.45)	ND
Romaine hearts	7.40	100 g	OLC 1056 *stx2h*	2.5	5.05 (1.1-23.1)	ND
Ground Beef	4.50	325 g	OLC 3602 *stx2i*	2.2	0.4 (0.05-2.9)	40

CFU: Colony forming units per analytical unit. Estimate of inoculum by plating on agar media.
MPN (CL): Estimate of STEC post-equilibration as Most Probable Number per analytical unit (95% confidence limits).
ND: Not Done

4.5 Recovery Of Injured Cells

CAM Vol 1. Annex 4.2 (*Supplement To The Procedure For The Development And Management Of Food Microbiological Methods*), requires that to assess analysis of foods that are processed under conditions which can generate sub-lethal injury to bacterial cells, that the food items should be inoculated with cells that have first been subjected to a stress that results in injury of 80% of inoculating cells.

In addition to pre-inoculation stress, the inoculating culture is also required to be added to the food item at levels low enough that not all analytical units contain the target (fractional positives). The inoculated cells must then be equilibrated in the food from 48 hours to 2 weeks, depending on whether the product is refrigerated, dried or frozen (CAM Vol 1. Part 4: *Guidelines For The Relative Validation Of Indirect Qualitative Food Microbiological Methods*. Section 4.8).

This experimental design confounds the influence on the probability of detection of three separate factors, the limit of detection, pre-inoculation cell injury, and loss of cell viability during equilibration. Consequently, a negative result from an analytical unit inoculated at low/fractional levels with injured cells may be the result of four different possibilities:

1. The volume of inoculum added to the analytical unit did not contain one or more viable cells;
2. Inoculated cells lost viability during the equilibration period;
3. The method fails to support the replication of injured cells in enrichment.
4. The method fails to detect the inoculated organism.

An experimental design in which injured cells are inoculated at fractional levels, does not allow these four possibilities to be differentiated, and only two of these possibilities are related to method performance. This issue is magnified for an absolute validation, as in a relative validation the alternate method can be compared to a reference method, allowing the proportion of positive analytical units to be compared.

Instead of using the standard experimental design that confounds cells injury, loss of viability during equilibration and the limit of detection, experiments were conducted to quantify the proportion of analytical units from which injured STEC cells can be recovered during enrichment.

Experiments to test the recovery of injured cells were conducted with one food item each from the Food Categories Dairy, Fruit and Vegetable Based Products, and Raw Meat (**Table. 14**). Commercially sourced Alfred le Fermier Grand Cru cheese (raw milk, hard texture), Romaine lettuce, and ground beef were used with analytical units of 25 g.

A suspension of the inoculating STEC strain of approximately 7.5 log CFU/mL was prepared in PBS as described above in Section 4.4. The cell suspension was then stressed by either heat treatment or exposure to lactic acid. Heat treatment consisted of adding 1 mL of cell suspension to 9 mL of PBS at 55 °C and incubating for 10 min, then 1 mL of the treated cells were diluted into 9 mL of PBS in an ice bath. Lactic acid stress consisted of adding 1 mL of cell suspension to 9 mL 1% DL-lactic acid and incubating for 15 min, then 1 mL of the treated cells were diluted into 9 mL of fresh PBS. An inoculum was prepared from the injured cells by diluting in PBS to achieve a concentration of less than 10 CFU/100 µL. The concentration of cell suspension was determined by plating dilutions on to duplicate TSA, incubated for 24 h at 35 °C. The cell injury level was determined by plating dilutions onto duplicate TSA (all cells) and MacConkey (MAC) (uninjured cells) agar incubated for 48 h at 35 °C. The concentration of the inoculum and proportion of analytical units inoculated with injured and uninjured cells was estimated by replicate plating onto twenty TSA and twenty MAC plates of 100 µL of the treated cell inoculum. These plates were incubated for 48 h at 35 °C.

The treated cells were then used to inoculate twenty analytical units of 25 g. A single analytical unit was inoculated with untreated cells as a positive control, and five uninoculated analytical units were retained as negative controls. All the analytical units were then analysed in parallel for STEC as described in Section 4.5 with the following modification. After 16 hours total incubation a 10 mL aliquot of the enrichment was taken and processed for *stx* screening and isolation, this process was repeated on the same enrichments at 24 hours total incubation. STX real-time PCR screening and isolation was attempted at both 16 hours and 24 hour enrichment to determine if there was a difference in STEC detection and isolation between these time points.

The background microbiota of the food item was determined by analysis of three uninoculated analytical units of 10 g for total aerobic plate count by MFHPB-18 (GoC, 2025).

Table. 14. Recovery of Injured Cells. Food item, microbiota, inoculated STEC strain, inoculum concentration, injury method and percentage injury. Proportion of analytical units STEC positive after 16 h or 24 h enrichment.

Food Item	Microbiota Log CFU/g	Strain	Inoculum CFU	Inoculum Stress	% Injury
Raw Milk Cheese	6.3	BMH-19-0006 *stx1a, 2a, 2d*	5.6	Lactic acid	>99.9%
Romaine hearts	5.2	11-3088 *stx2a, aaiC*	4.0	Heat	>99.9%
Ground Beef	3.8	1934 *stx1a, 2a, eae*	5.2	Heat	98.9%

% Injury: 1 minus (CFU on MacConkey agar divided by CFU on Tryptic Soy agar)

5. RESULTS

5.1 Method Performance: POD and Performance Parameters

Probability of Detection (POD) and performance parameters (Relative Sensitivity, False Negative, Relative Specificity, False Positive Rate, False Negative Rate and Test Efficacy) were calculated as described in CAM Vol 1. Annex 4.4 (*Procedure For The Statistical Evaluation And Calculation Of Performance Parameters Of A New Alternative Qualitative Method Compared To A Reference Cultural Method*). for enrichment screening by STX real-time PCR with DNA extracts prepared with **InstaGene Matrix** (**Table 15**) or **Puregene Cell Kit** (**Table 16**). POD was also determined for isolation of STEC on agar media (**Table 17, Table 18**). The data sheets for each Food Type are provided as **SD-08 to SD-16**. The POD analysis calculations are available as **SD-17**.

The overall performance parameters for STEC detection in *stx* positive enrichments, met the Compendium standard qualitative analysis for all parameters (Sensitivity ≥98%; Specificity ≥90.4%; False Negative Rate <2%; False Positive Rate <9.6%; Efficacy ≥ 94%) with both InstaGene Matrix and Puregene Cell Kit DNA extracts from enrichment broth (**Table 15, Table 16, SD-17 Tabs 1 and 2**). However, the performance of the screening using InstaGene Matrix did not achieve the Compendium standard with the Food Type Nuts/Nut Meats (**Table 15**) due to three False Negatives.

The validation trial protocol, following MFLP-52, recommend plating of enrichment dilutions 10^{-4}, 10^{-5} and 10^{-6} on both RBA and MAC-T agar. Under this protocol the overall performance parameters for STEC isolation from *stx* positive enrichment was below the compendium standard for two parameters Sensitivity (94.4% vs ≥98%) and False Negative rate (5.6% vs <2%) (**Table 17, SD-17 Tab 3**). This was due to a low rate of isolation from two Food Types Dairy – Raw (7 False Negatives) and Fresh Meats – Processed (6 False Negatives), which had been co-inoculated with generic *E. coli* at 10x the level of the inoculated STEC. However, the proportion of False Negatives with these two Food Types is significantly reduced if the results of isolation of *stx* positive colonies from MAC-T agar plated with the 10^{-1} enrichment dilution is included (**SD-10** and **SD-15**). If these results are interpreted as STEC isolation the performance parameters meet the Compendium Standard (Sensitivity 98.5%, Specificity 100%, False Negative Rate 1.5%, False Positive Rate 0.0%, and Efficacy 99.1%) (**Table 18, SD-17 Tab 4**).

Additional support for revising the enrichment dilutions to be plated on agar media arises from a review of which enrichment dilutions produced *stx* positive colony pools (**SD-18**). Of 263 STEC positive samples for which data on the isolation pool was available, Rainbow agar plated with dilution 10^{-5} was mostly likely to produce a *stx* positive pool (n = 122), with a lower rate of isolation from dilution 10^{-4} (n = 28) due to confluent colony growth, and from dilution 10^{-6} (n = 38) due to sparse colonies. The proportion of samples with isolation from MAC-T increased as the enrichment dilution decreased from 10^{-6} (n = 2), 10^{-5} (n = 78), 10^{-4} (n = 89). Though even at dilution 10^{-4} colonies on MAC-T were often noted as absent or sparse. The proportion of samples with isolation from MAC-T plated with dilution 10^{-1} was not high (n = 11) but these plates were only examined for isolation if there had been no isolation from dilutions 10^{-4}, 10^{-5} and 10^{-6}. For these reasons the final protocol has been modified to recommend that isolation be first attempted by plating dilutions 10^{-4}, 10^{-5} and 10^{-6} dilutions onto RBA, and dilutions 10^{-1}, 10^{-2} and 10^{-3} onto MAC-T.

Table 15. Performance parameters of **screening enrichment for *stx*** with STX real-time PCR with **InstaGene Matrix** (**SD-17 Tab 1**)**.**

Food Category	Food Type (Data Sheet)	Total	TP	TN	FP	FN	Sensitivity	Specificity	FN rate	FP Rate	Test Efficacy
Dairy	Fermented (**SD-08**)	50	24	26	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Heat Processed* (**SD-09**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw (**SD-10**)	50	36	14	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Fresh Fruits and Vegetables	Nuts/Nut Meats (**SD-11**)	50	25	22	0	3	89.3%	100.0%	10.7%	0.0%	94.0%
	Raw and Raw Processed* (**SD-12**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Juices (**SD-13**)	50	21	29	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Raw Meats	Unprocessed (**SD-14**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Processed* (**SD-15**)	50	36	14	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Frozen (**SD-16**)	50	33	17	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Totals		450	264	182	0	4	98.5%	100.0%	1.5%	0.0%	99.1%

TP: true positive; TN: true negative; FP: false positive; FN; false negative. * Food Type co-inoculated with generic *E. coli.*

Table 16. Performance parameters of **screening enrichment for *stx*** with STX real-time PCR with **Puregene Cell Kit** (**SD-17 Tab 2**).

Food Category	Food Type (Data Sheet)	Total	TP	TN	FP	FN	Sensitivity	Specificity	FN rate	FP Rate	Test Efficacy
Dairy	Fermented (**SD-08**)	50	24	26	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Heat Processed* (**SD-09**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw (**SD-10**)	50	36	14	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Fresh Fruits and Vegetables	Nuts/Nut Meats (**SD-11**)	50	28	22	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Raw and Raw Processed* (**SD-12**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Juices (**SD-13**)	50	21	29	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Raw Meats	Unprocessed (**SD-14**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Processed* (**SD-15**)	50	36	14	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Frozen (**SD-16**)	50	33	17	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Totals		450	267	182	0	1	99.6%	100.0%	0.4%	0.0%	99.8%

TP: true positive; TN: true negative; FP: false positive; FN; false negative. * Food Type co-inoculated with generic *E. coli.*

Table 17. Performance parameters of **isolation of STEC on agar media** (**SD-17 Tab 3**).

Food Category	Food Type (Data Sheet)	Total	TP	TN	FP	FN	Sensitivity	Specificity	FN rate	FP Rate	Test Efficacy
Dairy	Fermented (**SD-08**)	50	24	26	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Heat Processed (**SD-09**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw* (**SD-10**)	50	29	14	0	7	81.6%	100.0%	19.4%	0.0%	86.0%
Fresh Fruits and Vegetables	Nuts/Nut Meats (**SD-11**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw and Raw Processed* (**SD-12**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Juices (**SD-13**)	50	21	29	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Raw Meats	Unprocessed (**SD-14**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Processed* (**SD-15**)	50	30	14	0	6	83.3%	100.0%	16.7%	0.0%	88.0%
	Frozen (**SD-16**)	50	33	17	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Totals		450	253	182	0	15	94.4%	100.0%	5.6%	0.0%	96.7%

TP: true positive; TN: true negative; FP: false positive; FN; false negative. * Food Type co-inoculated with generic *E. coli.*

Table 18. Performance parameters including **isolation of STEC on agar media, including *stx* positive colonies from 10^{-1} enrichment dilution on MAC-T** (**SD-17 Tab 4**).

Food Category	Food Type (Data Sheet)	Total	TP	TN	FP	FN	Sensitivity	Specificity	FN rate	FP Rate	Test Efficacy
Dairy	Fermented (**SD-08**)	50	24	26	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Heat Processed* (**SD-09**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw (**SD-10**)	50	36	14	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Fresh Fruits and Vegetables	Nuts/Nut Meats (**SD-11**)	50	27	22	0	1	96.4%	100.0%	3.6%	0.0%	98.0%
	Raw and Raw Processed* (**SD-12**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Juices (**SD-13**)	50	21	29	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Raw Meats	Unprocessed (**SD-14**)	50	31	19	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
	Processed* (**SD-15**)	50	34	14	0	2	94.4%	100.0%	5.6%	0.0%	96.0%
	Frozen (**SD-16**)	50	33	17	0	0	100.0%	100.0%	0.0%	0.0%	100.0%
Totals		450	264	182	0	4	98.5%	100.0%	1.5%	0.0%	99.1%

TP: true positive; TN: true negative; FP: false positive; FN; false negative. * Food Type co-inoculated with generic *E. coli.*

5.2 Limit of Detection

The raw data for the LOD experiments is presented in **SD-19 LOD Data.xlsx (Tab 1. Raw Milk Cheese; Tab 2. Romaine Lettuce; Tab 3. Ground Beef**). The limit of detection was determined through the artificial inoculation of a food matrix representative of the food category (Raw milk cheese 100 g, Romaine lettuce 100 g, and Ground Beef 325 g) at a range of known inoculum concentrations and equilibrated for 3 days at 1-5 °C prior to analysis, as described in **Section 4.5**. Six replicate analytical units were analyzed for up to six inoculation levels. To enumerate the level of the target organism in the analytical units CAM Vol 1., Annex 4.5 requires that MPN of the target organism is conducted with a suitable reference method. However, in the absence of a reference method this study was conducted as an absolute validation and so MPN analytical units were analyzed with the method being validated. This creates a self referential result, in which an underestimation of the target organism numbers by MPN will make the LOD appear lower and the method more sensitive than it is. To address this short coming the inoculum was enumerated by replicate plating of the highest inoculating suspension. This gives a more precise figure than MPN, and since the food items were stored below 7 °C, the minimum growth temperature for *E. coli*, the inoculated STEC could not replicate between inoculation and analysis. MPN was calculated using the method of the FDA Bacteriological Analytical Manual (Blodgett, 2023). Comparison of the CFU and MPN values for all three food items indicated no change in STEC levels between inoculation and analysis, as the CFU values were within the 95% confidence limits of the MPN. LOD was estimated by the CAM Vol., Annex 4.5 method as the STEC level that lies between the two levels that give respectively more and less than 50% (3/6) positives (**Table 19**).

Table 19. Limit of Detection Experiments. Food Item, microbiota levels, analytical unit. Inoculated STEC, inoculation level and estimated inoculation levels above and below 50% positive.

	Analytical	Highest Inoculum		Proportion of Units STEC Positive						
	Unit (g)	CFU	MPN	CFU	+ve	CFU	+ve	CFU	+ve	LOD
Raw Milk Cheese	100 g	2.3	1.65 (0.45 – 5.45)	2.3	6/6	1.1	2/6	0.55	4/6	1.7
Romaine lettuce	100 g	2.5	5.05 (1.1 – 23.1)	2.5	5/6	1.30	5/6	0.63	2/6	0.97
Ground Beef	325 g	2.2	0.4 (0.05 – 2.9)	1.1	4/6	0.55	3/6	0.28	2/6	0.55

+ve: number of analytical units positive for STEC
CFU: Colony forming units per analytical unit. Estimate of inoculum by plating on agar media.
MPN (CL): Estimate of STEC post-equilibration as Most Probable Number per analytical unit (95% confidence limits).

LOD 50% and LOD 95% was also calculated from the same data by the method of Wilrich and Wilrich (2009) (**Table 20**). An example calculation sheet is available as **SD-20 PODLOD ver10A 325 g beef.xls**.

Table 20. Limit of Detection CFU/analytical unit, 50% probability of detection and 95% probability of detection in an analytical unit.

Matrix	Matrix effect F_i	Log matrix effect f_i	SD of log matrix effect s_{fi}	Limit of Detection CFU		Test statistic Matrix effect Z_i
				50% (CL)	95% (CL)	
Raw Milk Cheese	1.014	0.014	0.322	0.68 (0.36 – 1.30)	2.95 (1.55 – 5.62)	0.044
Romaine lettuce	1.152	0.141	0.357	0.60 (0.29 – 1.23)	2.60 (1.27 – 5.32)	0.401
Ground Beef	1.167	0.154	0.349	0.59 (0.30 – 1.20)	2.57 (1.28 – 5.16)	0.424

(CL): 95% confidence limits
CFU: Colony forming units per analytical unit.

For all three food items the LOD was determined to approach 1 CFU per analytical unit, with analytical units of 100 g for raw milk cheese and Romaine iettuce, and 325 g for ground beef.

5.3 Recovery of Injured Cells

The raw data for Recovery of Injured Cells presented in **SD-21 Cell Injury Data.xlsx** (**Tab 1. Raw Milk Cheese; Tab 2. Romaine Lettuce; Tab 3. Ground Beef**). The ability of the method to recover injured cells was assessed by POD analysis (**SD-22 Tab 1**) and Chi square analysis (**SD-22 Tab 2**) to determine if there was a significant difference in the probability of recovery of STEC between direct plating of injured cells onto TSA, and recovery of STEC from the inoculated food items by isolation with the revised STEC Method. Since no difference in *stx* screening results were observed between the two DNA extraction methods (InstaGene Matrix and Puregene Cell Kit), analysis was compared only between 16 and 24 hour enrichment periods (**Table 21**).

Table 21. Recovery of injured STEC cells from food items.

Food	Injury	Recovery on agar		Enrichment Period	Recovery with STEC Method	dPOD (CL)	Chi test p-value
		TSA	MAC				
Ground beef	Heat	20	1	16 hours	18	0.10 (-0.04 - 0.26)	0.147
				24 hours	18	0.10 (-0.04 - 0.26)	0.147
Romaine Lettuce	Heat	20	0	16 hours	14	0.30* (0.12 – 0.48)	0.008*
				24 hours	15	0.25* (0.08 – 043)	0.017*
Raw Milk Cheese	Lactic	20	0	16 hours	19	0.05 (-0.08 - 0.20)	0.311
				24 hours	19	0.05 (-0.08 - 0.20)	0.311

TSA: Tryptic Soy agar
MAC: MacConkey agar
dPOD (CL): difference in probability of detection (95% confidence limits).
*: Statistically significant difference between replicate plates positive for injured cells and STEC Method
Heat: Cells suspended in phosphate buffered saline at 55 °C for 10 min
Lactic: Cells suspended in phosphate buffered saline with 1% lactic acid for 15 min.

The recovery rate of lactic acid injured STEC from Raw Milk Cheese after 16 hours and 24 hours enrichment was 95% (19/20) which was not different from recovery of TSA as determined by difference in probability of detection (dPOD) (**SD-22 Tab 1**) and Chi-squared test from a 2x2 contingency table (**SD-22 Tab 2**). The recovery rate of heat injured STEC from Ground beef after 16 hours and 24 hours enrichment was 90% (18/20), which was not statistically different from recovery of TSA (**SD-22 Tab 1 and Tab2**). The recovery rate of heat injured STEC from Romaine Lettuce was 70% (14/20) after 16 hours enrichment, and 75% (15/20) after 24 hours enrichment. Recovery of heat injured cells from lettuce at both time periods was determined to be significantly different than recovery on TSA, by both dPOD (**SD-22 Tab 1**) and Chi-squared test from a 2x2 contingency table (**SD-22 Tab 2**).

6. SUMMARY

6.1 Inclusivity and Exclusivity of PCR

The inclusivity and exclusivity of the STX real-time PCR was assessed against cultures of 160 *stx* positive strains and 70 *stx* negative strains. All strains provided the expected result, indicating a Sensitivity of 100%, a Specificity of 100%, a False Negative Rate 0%, and a False Positive Rate 0% **(Table 2)**. The specificity of the STX real-time PCR primers and probes was confirmed through *in silico* analysis of greater than 30,000 bacterial genome sequences. No false positive or false negative results were determined *in silico*.

The STX real-time PCR was determined to successfully detect the colonies of *stx* positive organisms plated on six agar media. BHI (n = 70 strains), MacConkey (n= 68 strains), RBA (n = 24 strains), MAC-T (n = 24 strains), TSA (n = 24 strains), and Nutrient agar (n = 24 strains). These results support the use of the STX real-time PCR for the characterisation of colonies from plates of these six media.

The reliability of the STX real-time PCR was demonstrated by the absence of false positive results and only one false negative result out of 82 positive reactions in analysis of the same DNA extracts by five laboratories using four real-time PCR platforms (Lightcycler 96; CFX96; Quantstudio 5 and 6). The overall performance statistics for the four real-time PCR platforms was Sensitivity 98.8%; Specificity 100%; False Negative Rate 1.2%; False Positive Rate 0.0%; Agreement 99.2% (**Table 4**).

Inclusivity and exclusivity of the CF real-time PCR was assessed against cultures of 108 strains positive for *eae*, *aggR* or *aaiC* and 72 strains negative for these three genes. All strains provided the expected result, indicating a Sensitivity of 100%, a Specificity of 100%, a False Negative Rate 0%, and a False Positive Rate 0% (Table 5). The specificity of the CF real-time PCR primers and probes was confirmed through *in silico* analysis of greater than 30,000 bacterial genome sequences. No false positive results were determined. The CF real-time PCR was determined to detect all known *eae* variants except *eae*-citrobacter. However, *eae*-citrobacter has only been reported in isolates of *Citrobacter* species, a non-target organism.

The CF real-time PCR was determined to successfully detect colonies of strains carrying *eae*, *aggR* and/or *aaiC* plated on six agar media. BHI (n = 47 strains), MacConkey (n= 61 strains), RBA (n = 22 strains), MAC-T (n = 22 strains), TSA (n = 22 strains), and Nutrient agar (n = 22 strains). These results support the use of the CF real-time PCR for the characterisation of colonies from plates of these six media.

The reliability of the CF real-time PCR was demonstrated by the absence of false positive or negative results from analysis of the same DNA extracts using three real-time PCR platforms (Lightcycler 96; CFX96; Quantstudio 6).

The results of this study demonstrate that the STX real-time PCR and the CF real-time PCR are highly specific for the target DNA sequences, can be implemented on multiple platforms, and can be used successfully with colonies grown on six different agar media.

6.2 Performance Analysis

Screening of enrichment broths for STEC with the STX real-time PCR was very reliable. DNA extracts prepared from the enrichment with both InstaGene Matrix and Puregene Cell Kit met the Compendium standard for qualitative analysis for all parameters (Sensitivity ≥98%; Specificity ≥90.4%; False Negative Rate <2%; False Positive Rate <9.6%; Efficacy ≥94%). A single false negative (Dairy – Heat Treated) STX real-time PCR result was observed with PureGene DNA extracts (Sensitivity 99.6%; Specificity 100%; False Negative Rate 0.4%; False Positive Rate 0.0%; Efficacy 99.8%) (**Table 16**). Detection of *stx* in enrichments with InstaGene Matrix also met the Compendium criteria (Sensitivity 98.5%; Specificity 100%; False Negative Rate 1.5%; False Positive Rate

0.0%; Efficacy 99.1%), but was less reliable with four false negative results (**Table 15**). The higher rate of false negative results occurred with Nuts/Nut Meats (1 Dairy – Heat Treated and 3 Fresh Fruits and Vegetables – Nuts/Nut Meats) indicating that InstaGene Matrix is less suitable than PureGene for this Food Type.

The InstaGene Matrix protocol is faster and has less steps than the Puregene Cell Kit. The results of the validation trial confirm that InstaGene Matrix is suitable for routine analysis of a wide range of food items, but also indicate that the use of the Puregene Cell kit for DNA extraction can enhance the performance of the method when PCR inhibitors are present in the sample matrix.

The adoption of a minimum enrichment period of 16 hours and a maximum of 24 hours, is supported by method performance data with enrichments of 18 to 24 hours (**Tables 15, 16, 17, 18**) and by the cell injury recovery experiments, in which no difference in the detection and isolation of STEC from food inoculated with injured cells was observed between aliquots of the same enrichments analysed after 16 or 24 hour incubation (**Table 21**).

Isolation of STEC from positive enrichments remains challenging with some samples. The overall performance parameters for STEC isolation from *stx* positive enrichment was below the compendium standard for two parameters Sensitivity (94.4% vs ≥98%) and False Negative rate (5.6% vs <2%) (**Table 17**). However, if the results of isolation of *stx* positive colonies from MAC-T agar plated with the 10^{-1} enrichment dilution are interpreted as STEC isolation the performance parameters meet the Compendium Standard (Sensitivity 98.5%, Specificity 100%, False Negative Rate 1.5%, False Positive Rate 0.0%, and Efficacy 99.1%)(**Table 18**). For this reason, the final protocol was modified to recommend the plating on MAC-T of a lower range of enrichment dilutions. False negative results in STEC isolation can be attributed to the low probability of selecting STEC colonies from agar plates when other bacteria compose the majority of colonies on the isolation media. In support of this hypothesis, the highest rate of false negative samples occurred with the food types Dairy – Raw and Raw Meats – Processed, which in the experiments were inoculated with generic *E. coli* at ten times the level of inoculated STEC. However, the rate of false negatives was reduced when isolation was supported by plating low dilution for enrichment media on to MAC-T, which inhibits colony formation by a larger proportion of generic *E. coli*, than it does STEC (**SD-02**). Though this strategy will not work for all STEC strains as 42% were observed to be sensitive to MAC-T.

For routine analysis, this protocol recommends screening a minimum of 60 colonies (6 pools), before declaring a negative result. The validation was conducted using a maximum of 60 colonies to reflect typical routine conditions. The enrichment screening by STX real time PCR was highly reliable for the detection of STEC, with no false positive results observed in any of the experiments. Consequently, failure to isolate STEC from 60 colonies from *stx* positive enrichment most likely indicates that the STEC are present but are outnumbered by other organisms on the agar plates. Consequently, when the method is used for analysis of outbreak associated samples, or in similar priority situations and the enrichment is positive for *stx*, testing additional colony pools is recommended.

6.3 Limit of Detection

LOD was assessed with analytical units of 100 g for raw milk cheese, 100 g for Romaine lettuce and 325 g for ground beef. The LOD 50% for all three food items were 0.68 CFU for cheese, 0.60 for lettuce, 0.59 for ground beef (**Table 20**). These values approach the theoretical maximum for sensitivity of 1 CFU per analytical unit. These results provide evidence that the same LOD is achievable with lower analytical unit sizes and analytical units up to 100 g for Dairy Products and Fresh Fruits and Vegetables, or up to 325 g for Raw Meats.

6.4 Recovery of Injured Cells

No significant difference between the probability of recovery of heat or lactic acid injured STEC cells on ground beef or raw milk cheese was found by POD analysis or Chi-squared analysis. No significant difference was observed between recovery of injured cells from enrichments incubated between 16 and 24. These results validate the protocol for an enrichment period minimum of 16 and a maximum of 24 hours.

Recovery of heat injured cells was determined to be significantly lower by POD analysis for cells on Romaine lettuce, with 70% of analytical units positive at 16 h, and 75% positive at 24 h. These results indicate that some food matrices may inhibit the recovery of injured cells, but that confidence can be placed on the recovery of heat and lactic acid injured STEC cells from meats and dairy products, but there may be a lower rate of recovery of injured cells from other matrices, including leafy greens. Leafy greens were selected for these experiments as STEC and leafy greens are considered a high risk food pathogen combination, though STEC in these products are unlikely to experience heat stress.

6.5 Conclusions

Due to the changes in enrichment conditions from MFLP-52, and the expansion of the range of *stx* variants detected by this method it was not possible to conduct a parallel validation study with MFLP-52. After discussion with the Health Canada and CFIA Microbiological Methods Committee it was decided to proceed with the absolute validation study described in this report.

The results presented in this report indicate that STEC detection by STX real-time PCR of enrichment broth is robust, reliable, highly sensitive, and inclusive of the nineteen described Stx subtypes. The new protocol offers clear improvements in analytical unit size, range of STEC detected, and the recovery of injured STEC compared to MFLP-52.

Post-enrichment isolation of STEC remains potentially challenging when other microbiota form the predominant number of colonies on isolation media. There is no ideal solution to this challenge as there are no selective agents or conditions that are unique to STEC, that are not shared with other related bacteria. Similarly, the potential to produce Stx remains the only differential characteristic known to be unique to STEC. However, the results of this study do indicate that the rate of STEC isolation from foods can be improved by attempting isolation in parallel on low selectivity RBA and high selectivity MAC-T media.

The authors of this report are confident that these results support the replacement of MFLP-52 with this protocol for the analysis of food samples for STEC.

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8. List of Supporting Documents.

SD-01 Recovery in Enrichment of Injured Cells.xlsx
SD-02 Growth of E coli on MacConkey with tellurite.xlsx
SD-03 STX PCR and CF PCR Inclusivity Exclusivity Results.xlsx
SD-04 STX rtPCR in silico inclusivity and exclusivity.xlsx
SD-05 STX PCR inter-laboratory study.xlsx
SD-06 CF rtPCR in silico inclusivity and exclusivity.xlsx
SD-07 CF rtPCR different platforms.xlsx
SD-08 Dairy - fermented.xlsx
SD-09 Dairy - Heat Processed.xlsx
SD-10 Dairy - Raw.xlsx
SD-11 F&V - Nuts and nut meats.xlsx
SD-12 F&V - Raw and raw processed.xlsx
SD-13 F&V - Juices.xlsx
SD-14 Raw Meats - Unprocessed.xlsx
SD-15 Raw meat - Processed.xlsx
SD-16 Raw meat - Frozen.xlsx
SD-17 POD analysis.xlsx
SD-18 Distribution of enrichment dilutions providing isolates.xlsx
SD-19 LOD Data.xlsx
SD-20 PODLOD_ver10a 325 g beef.xls
SD-21 Cell Injury Data.xlsx
SD-22 Cell Injury Statistical Analysis.xlsx